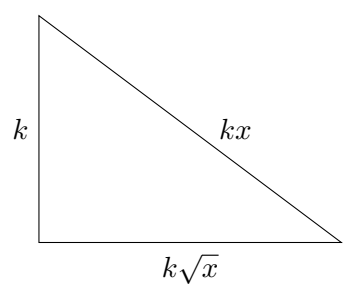
$A, B, S \in \mathcal{C}$ and $f : A \to B$

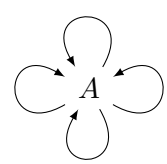
$$- \times S = \begin{cases} A \times S & \# \ A \in \mathcal{C} \\ f \times \mathrm{id}_S : A \times S \to B \times S & \# \ (a, s) \mapsto (f(a), s) \end{cases}$$

$$\begin{tikzcd} A \arrow[r, shift left, "f_1"] \arrow[r, shift right, "f_2"'] & B \arrow[r, "g"] & C \end{tikzcd}$$

For sets a mapping $g : B \to C$ is said to be *injective* if for $b_1, b_2 \in B$ we have $g(b_1) = g(b_2) \Rightarrow b_1 = b_2$.

Not a good definition because it looks inside the objects. Rather, in category theory we define a more general concept, *monomorphism*. A morphism $g : B \to C$ is said to be a monomorphism if for all $f_i : A \to B$ we have $gf_1 = gf_2 \Rightarrow f_1 = f_2$.





$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "g \circ f"'] & B \arrow[dl, "g"] \\ C & \end{tikzcd}$$

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\eta_A"'] & F(B) \arrow[d, "\eta_B"] \\
G(A) \arrow[r, "G(f)"'] & G(B)
\end{tikzcd}$$

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\eta_A"'] & F(B) \arrow[d, "\eta_B"] \\
G(A) \arrow[r, "G(f)"'] & G(B)
\end{tikzcd}$$

$$G(\text{🦇}) \xrightarrow{n=\langle \text{💀}, \text{👻} \rangle} \text{👻}\ \text{🎃}_{i=1}^{n} H_i(\text{🧹})$$

Figure 1: Happy Halloween!

$$A \xrightarrow[abc\ldots z]{f_1+\cdots+f_n} \text{🧹} B \xrightarrow{f_1+\cdots+f_n} \text{🧹} C \xrightarrow[abc\ldots z]{} \text{🧹} D$$

Figure 2: Happy Halloween!

🧹🎃💀☁️ ➢ 👻🦇

0.6249999121401877

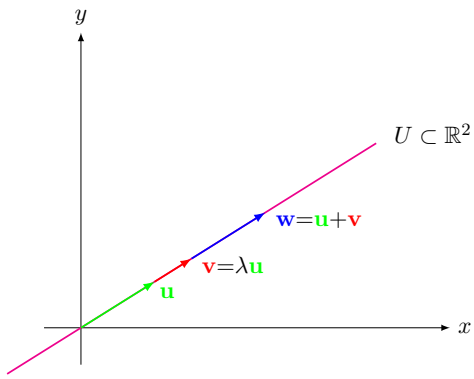


Figure 3: The line U is a linear subspace of $\mathbb{R}^2$

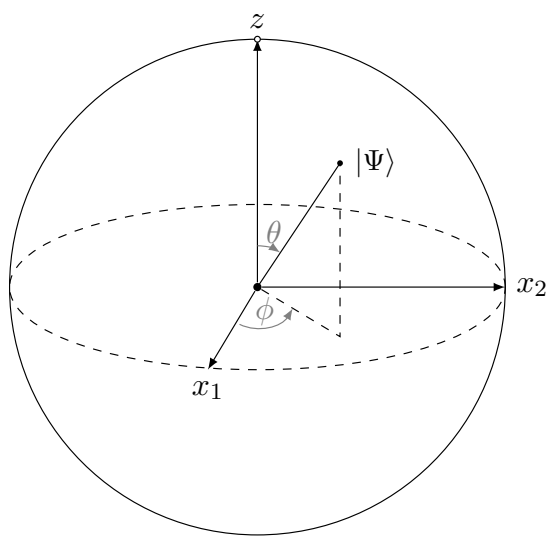


Figure 4: The Bloch Sphere

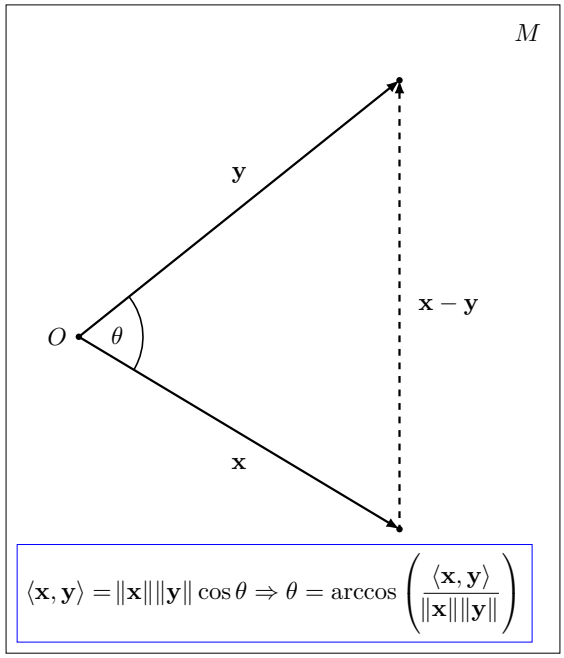


Figure 5: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

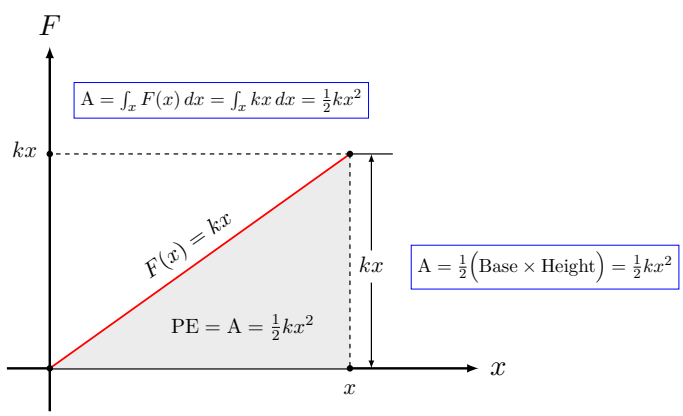


Figure 6: Potential Energy of a Simple Harmonic Oscillator

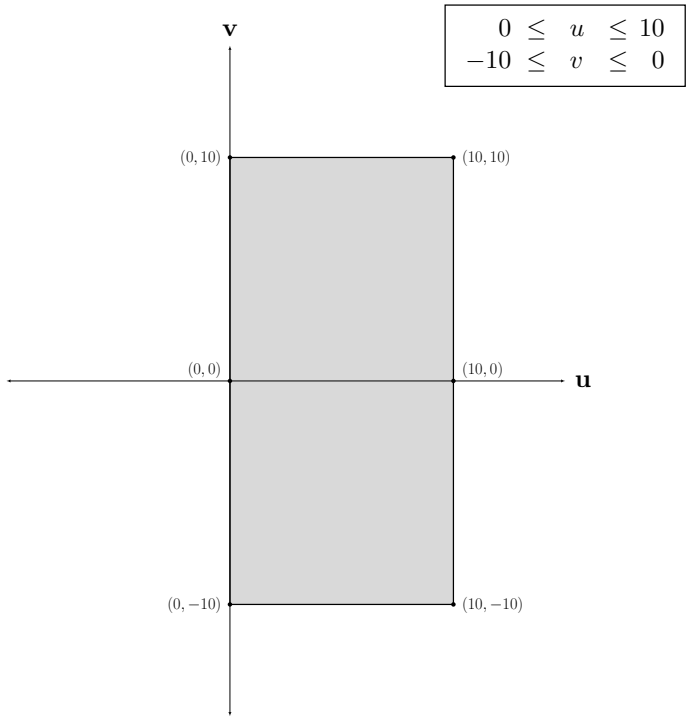


Figure 7: The transformed parallelogram is a rectangle in the uv-plane

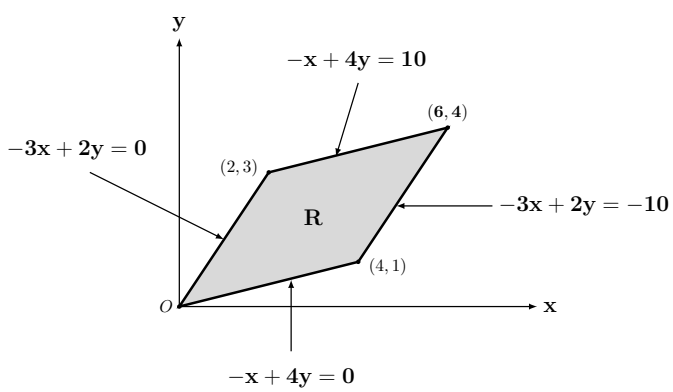


Figure 8: Equations of the Sides of the Parallelogram

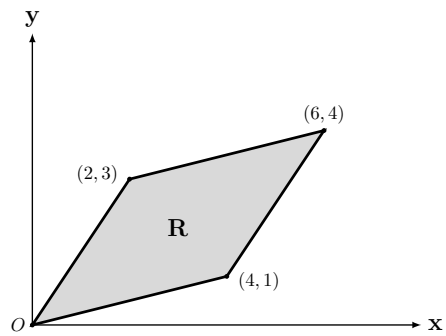


Figure 9: Parallelogram in the xy-plane

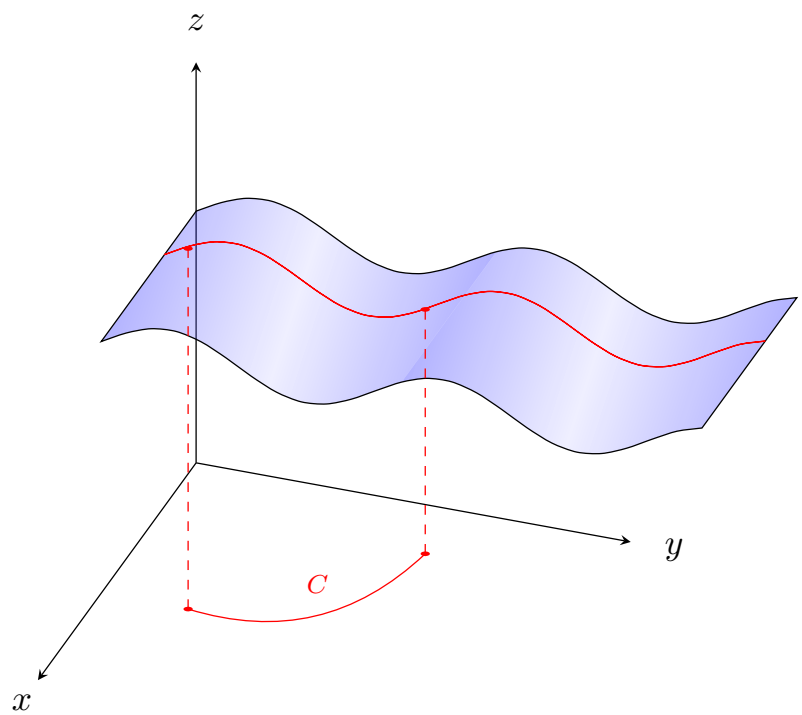


Figure 10: 3D

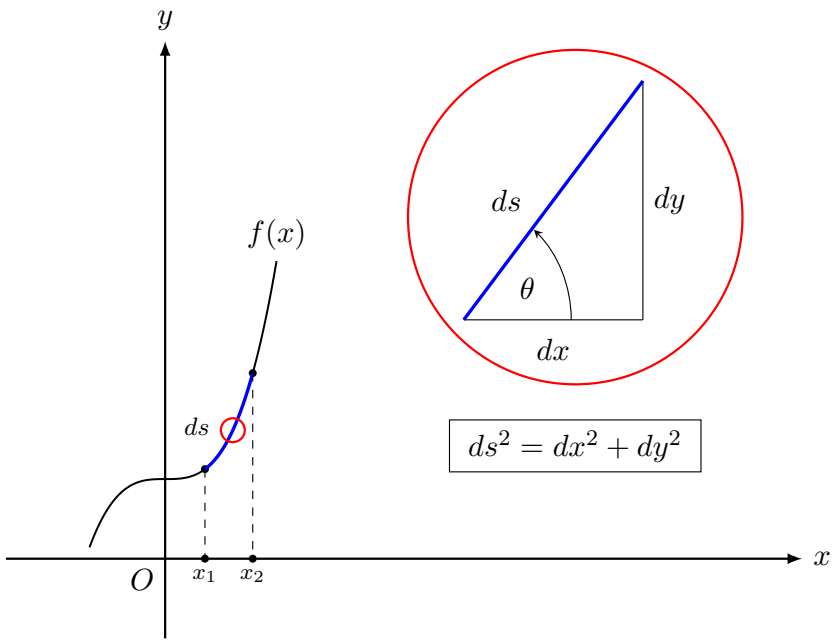


Figure 11: $f(x)$, ds and the Pythagorean Theorem

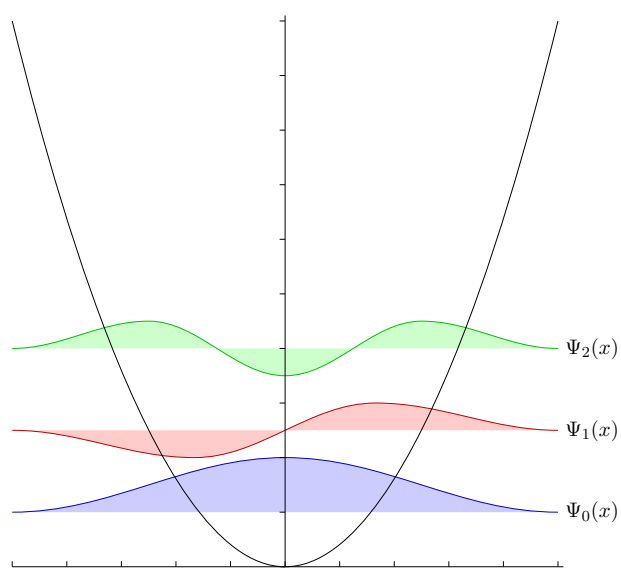


Figure 12: Wave Functions

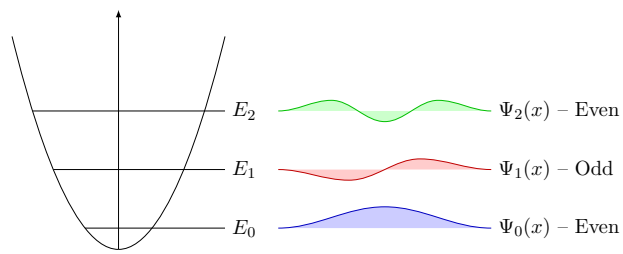


Figure 13: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	9
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 14: A Few Example Kaprekar Numbers

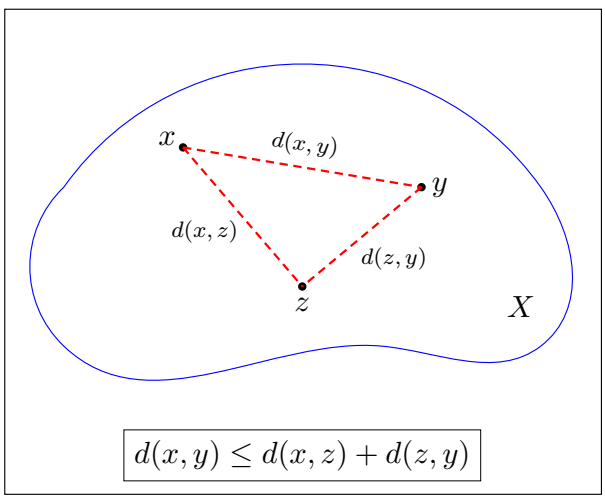


Figure 15: The Triangle Inequality

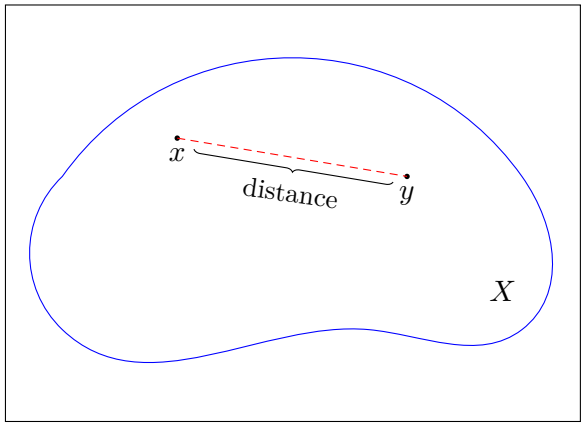


Figure 16: X, x and y

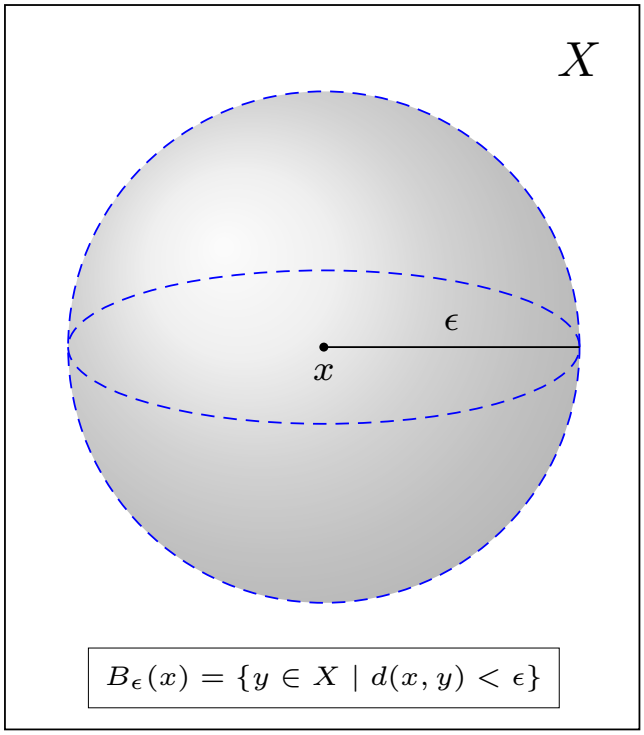


Figure 17: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

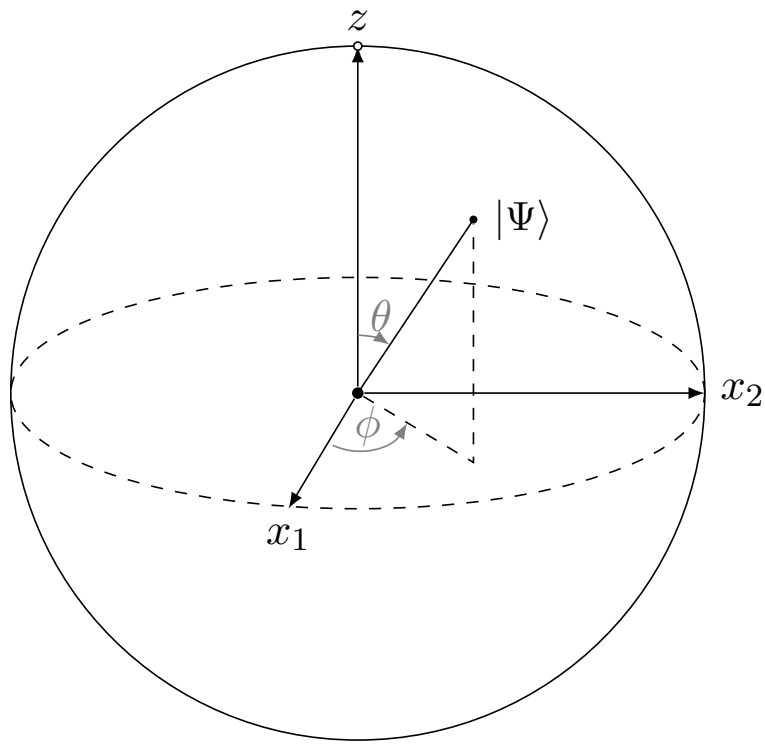


Figure 18: The Bloch Sphere

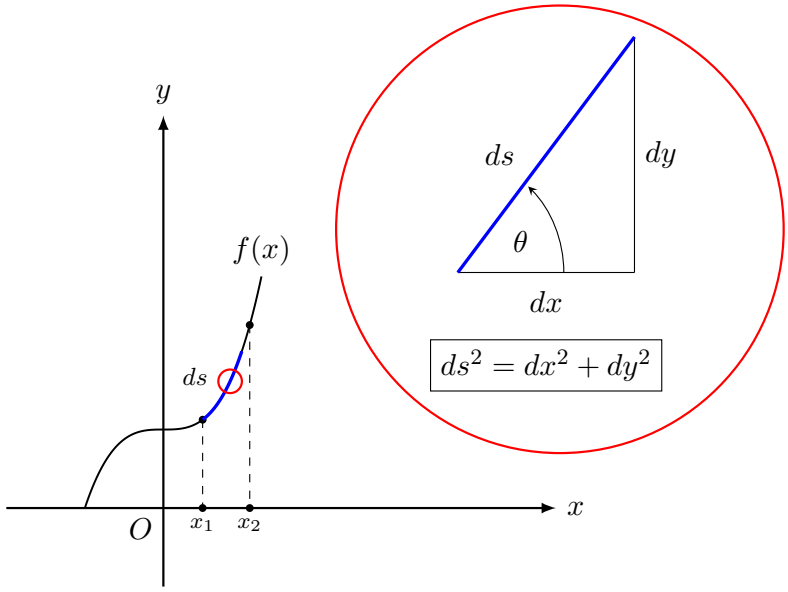


Figure 19: $f(x)$, ds and the Pythagorean Theorem

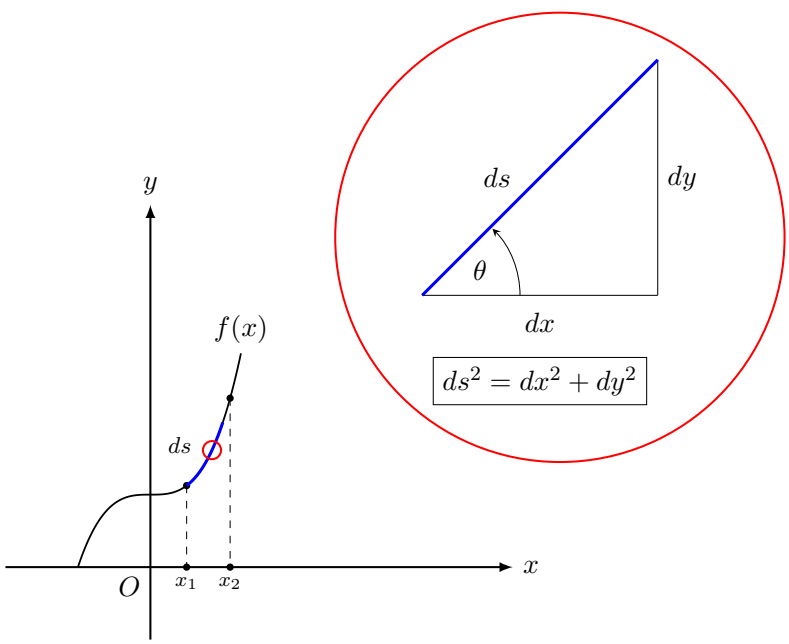


Figure 20: $f(x)$, ds and the Pythagorean Theorem